WorldQuant University
Master of Science in Financial Engineering

MScFE 622 — Stochastic Modeling

Group Work Project #2

Regime-Based ETF Rotation Using Markov Chains and Hidden Markov Models on VIX Dynamics

Asset Universe:	Team A: David Olutunde Daniel, B: Thao Nguyen Ta, C: (S ANIRUDDH)
Regime Driver:	CBOE Volatility Index (VIX)
Sample Period:	19 November 2004 – 30 December 2024
Observations:	5,061 trading days
Data Source:	Yahoo Finance

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Contents

1. Introduction

Regime-switching models have long been used in quantitative finance [1] to capture the well-documented stylised facts of financial returns: volatility clustering, fat tails, persistent bear and bull phases, and asymmetric correlations during crises [2, 3]. The underlying premise is that market dynamics are not stationary — they are governed by a small number of unobservable states with distinct risk–return characteristics, and the transitions between these states follow a Markov process.

The CBOE Volatility Index (VIX) is widely used as a proxy for the latent volatility state of the equity market because it is forward-looking (implied from S&P 500 options) and tends to spike sharply during periods of stress [4]. Conditioning asset allocation on the VIX regime is therefore a natural and economically interpretable mechanism for tactical rotation [5].

This project applies that framework to a three-asset universe representing the classical risk-on / risk-off / inflation-hedge triangle:

- **SPY** — SPDR S&P 500 ETF: the canonical risk-on asset, expected to outperform in calm and trending markets.
- **TLT** — iShares 20+ Year Treasury Bond ETF: the classic risk-off asset, expected to benefit from flight-to-quality flows and falling rates in crisis regimes.
- **GLD** — SPDR Gold Shares ETF: a real-asset hedge that tends to perform well during inflationary stress and prolonged equity weakness.

The remainder of the report follows the five-step structure mandated by the project brief: data preparation (§2), regime modelling (§3), model selection and interpretation (§4), strategy design (§5), and back-testing with sensitivity analysis (§6). The technical and non-technical conclusions (§7 and §8) translate the statistical findings into actionable investment guidance.

2. Step 1 — Data Preparation and Exploration

2.1 Data Sources and Sample

Daily adjusted closing prices for SPY, TLT, GLD, and the level of the VIX were downloaded from Yahoo Finance [6]. The intersection of the four series yields a maximum common sample of **5,061 trading days** spanning **19 November 2004 to 30 December 2024**, comfortably covering the 2008 Global Financial Crisis, the 2011 Eurozone crisis, the 2015–2016 emerging-market sell-off, the February 2018 “volmageddon” episode, the COVID-19 shock of March 2020, the 2022 inflation/rates regime, and the post-pandemic equity recovery.

2.2 Return Definitions

For each ETF $i \in \{\mathrm{SPY}, \mathrm{TLT}, \mathrm{GLD}\}$ the daily log-return is computed as

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \tag{1}$$

and the daily change in the volatility index is defined as

$$\Delta\mathrm{VIX}_t = \mathrm{VIX}_t - \mathrm{VIX}_{t-1}. \tag{2}$$

The signal ΔVIX is preferred to log-VIX returns because VIX is bounded below by zero and exhibits strong mean-reversion in levels; first differences provide a more symmetric and tractable process for regime inference.

2.3 Descriptive Statistics

Table 1 reports the unconditional descriptive statistics. All three ETF return distributions display the classical financial-return signatures: very low mean, fat tails, and pronounced negative skewness. SPY is the most volatile (annualised $\sigma \approx 19.0\%$), followed by GLD ($\approx 17.5\%$) and TLT ($\approx 14.8\%$). The min/max values reveal the magnitude of the COVID-19 March 2020 shock (SPY -11.6%, $+13.6\%$ in a single day).

Table 1: Descriptive statistics of daily ETF log-returns (2004–2024).

Statistic	SPY	TLT	GLD
Observations	5,061	5,061	5,061
Mean	0.000391	0.000124	0.000334
Std. Dev.	0.011976	0.009307	0.011055
Min	−0.1159	−0.0690	−0.0919
Q1	−0.0040	−0.0054	−0.0051
Median	0.000704	0.000441	0.000540
Q3	0.005781	0.005554	0.006098
Max	0.1356	0.0725	0.1070

Over the same period, the VIX averaged **19.1**, with a minimum of **9.1** (mid-2017 "low-volatility regime") and a maximum of **82.7** (16 March 2020, peak COVID panic).

2.4 Exploratory Visualisation

Figure 1 displays the three ETF log-return series. The familiar volatility-clustering pattern is unmistakable: large negative and positive moves cluster around the 2008–2009 GFC and the March 2020 COVID episode, while the periods 2013–2018 and 2023–2024 appear comparatively tranquil.

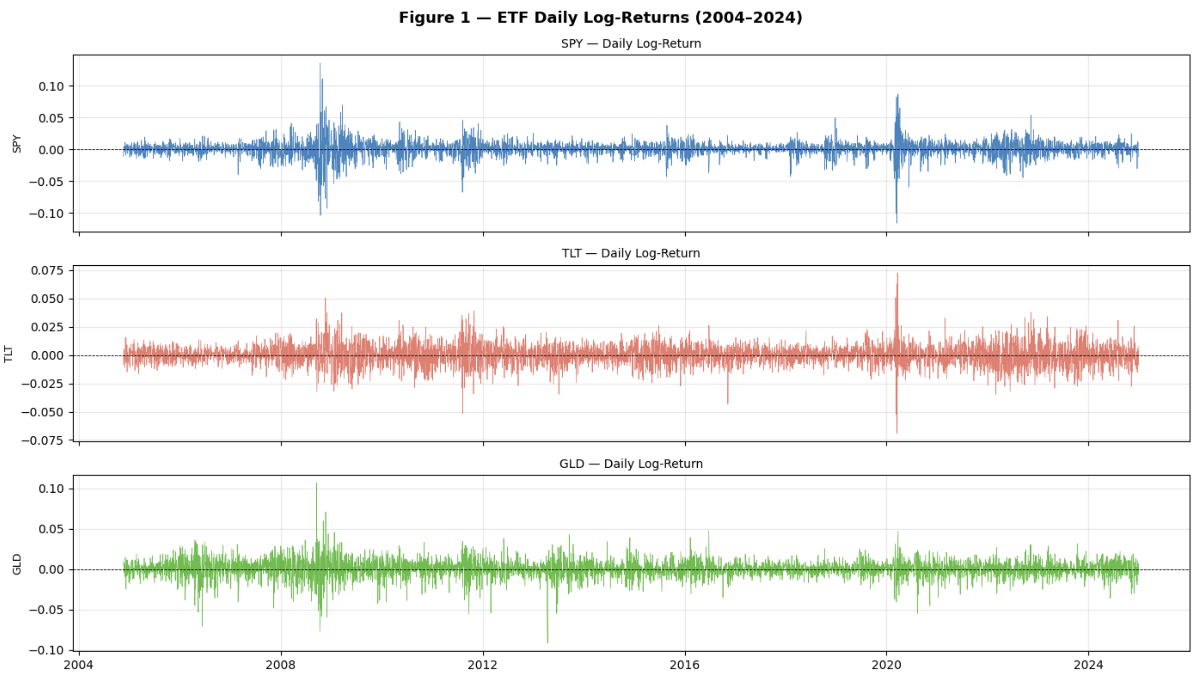

Figure 1: Daily log-returns of SPY (top), TLT (middle) and GLD (bottom), 2004–2024.

Figure 2 plots the level and first difference of the VIX. The level series is dominated by sharp upward spikes followed by slower mean-reverting decays — consistent with a multi-regime data generating process rather than a single Gaussian distribution.

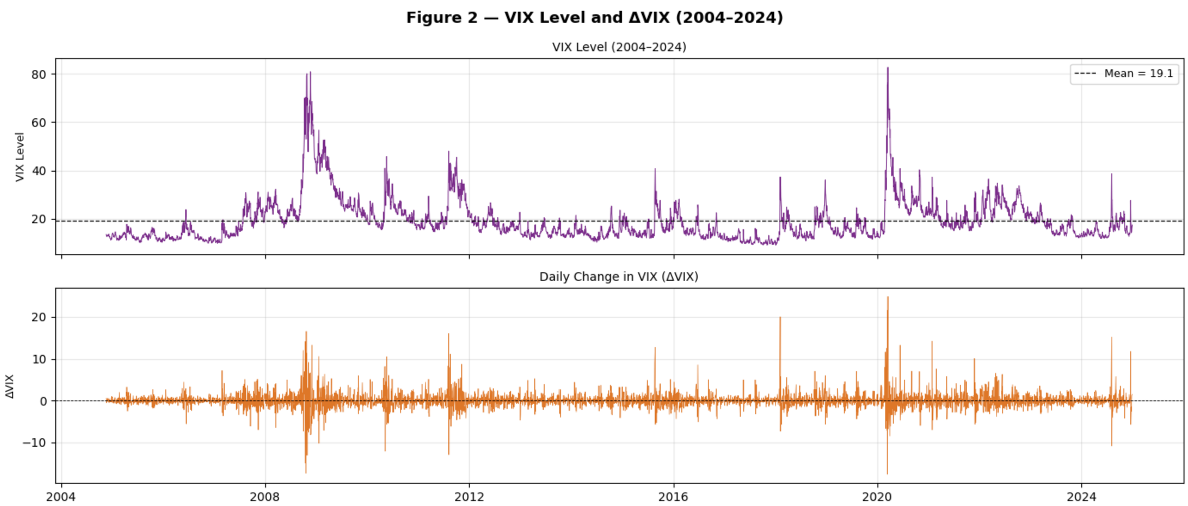

Figure 2: VIX level (top) and daily change ΔVIX (bottom), 2004–2024.

These two figures together confirm the central motivation of the project: return behaviour and volatility behaviour are visibly non-stationary, and both exhibit common turning points, suggesting that a hidden state variable indexing the volatility regime can usefully inform the allocation decision.

3. Step 2 — Modelling VIX Regimes

We estimate two families of regime models on the ΔVIX series: a discrete Markov chain on quantile-discretised states, and Gaussian Hidden Markov Models with continuous emissions.

3.1 Discrete Markov Chain (3-state)

ΔVIX is partitioned into three states by terciles of the empirical distribution: *Low*, *Medium*, and *High* ΔVIX. The maximum-likelihood estimate of the transition matrix P is obtained from observed state-pair counts:

$$\hat{P}_{\mathrm{MC}} = \begin{pmatrix} 0.3507 & 0.3010 & 0.3483 \\ 0.2383 & 0.4332 & 0.3286 \\ 0.4134 & 0.2639 & 0.3227 \end{pmatrix}. \tag{3}$$

The stationary distribution, computed as the left eigenvector of $\hat{P}_{\mathrm{MC}}$ corresponding to eigenvalue 1, is $\boldsymbol{\pi}_{\mathrm{MC}} = (0.3342,\ 0.3326,\ 0.3332)$ — essentially uniform, which is mechanically expected when states are defined by terciles.

Although the MC achieves a high in-sample log-likelihood (LL $= -5{,}478.08$) the diagonal probabilities are only ≈ 0.33–0.43. Persistence is therefore weak and the chain re-shuffles between states almost daily, which is undesirable for a tactical allocation rule (it would generate excessive turnover and short-lived regime calls).

3.2 Gaussian Hidden Markov Models

A more flexible specification treats the volatility state z_t as *latent* and assumes $\Delta\mathrm{VIX}_t \mid z_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2)$ with hidden Markov dynamics:

$$P(z_t = j \mid z_{t-1} = i) = A_{ij}, \qquad \sum_{j=1}^{K} A_{ij} = 1. \tag{4}$$

Parameters $\{\mu_k, \sigma_k, A_{ij}, \pi_i\}$ are estimated by the Baum–Welch [7,9] (Expectation–Maximisation [8]) algorithm.

2-State HMM.

$$\boldsymbol{\mu} = (-0.074,\ 0.214), \quad \boldsymbol{\sigma} = (0.805,\ 3.441),$$
$$A = \begin{pmatrix} 0.9613 & 0.0387 \\ 0.1100 & 0.8900 \end{pmatrix},$$

producing LL $= -8{,}560.91$, AIC $= 17{,}135.82$, BIC $= 17{,}181.52$.

3-State HMM.

$$\boldsymbol{\mu} = (-0.060,\ -0.016,\ 0.593),$$
$$\boldsymbol{\sigma} = (0.606,\ 1.698,\ 5.746),$$
$$A = \begin{pmatrix} 0.9413 & 0.0579 & 0.0008 \\ 0.0697 & 0.9126 & 0.0177 \\ 0.0000 & 0.1237 & 0.8763 \end{pmatrix},$$

producing LL = −8,239.35, AIC = 16,506.71, BIC = 16,598.12.

The 3-state HMM exhibits strong on-diagonal persistence (each state has $A_{ii} > 0.87$), and the three states are clearly ordered by ΔVIX volatility ($\sigma_1 < \sigma_2 < \sigma_3$), consistent with a Low / Medium / High Volatility regime interpretation.

Figure 3 is the Step 2 deliverable: the VIX level colour-coded by the decoded 3-state HMM sequence (Viterbi path [10], top panel) together with the smoothed posterior state probabilities $P(z_t = k \mid x_{1:T})$ (bottom panel). The Low (green), Medium (amber) and High (red) regimes align convincingly with well-known historical episodes: the GFC of 2008–2009, the European sovereign crisis of 2010–2011, the August 2015 / February 2018 vol spikes, and the March 2020 COVID-19 collapse.

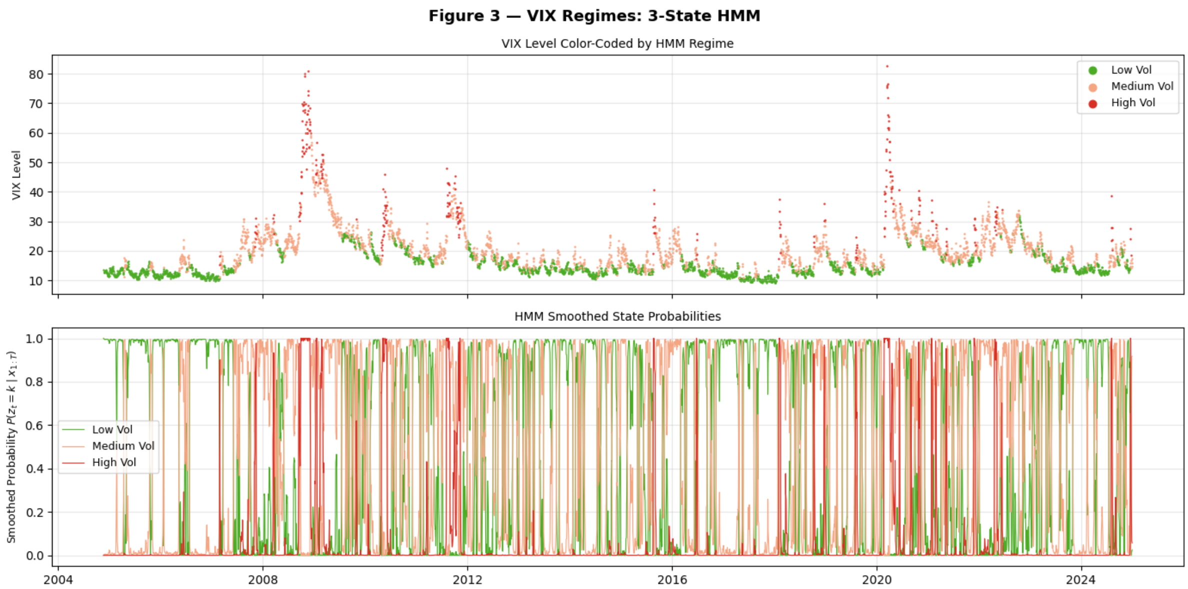

Figure 3: VIX level colour-coded by decoded HMM state (top) and smoothed posterior state probabilities (bottom). This is the Step 2 deliverable: a plot of VIX with colour-coded states.

4. Step 3 — State Selection and Interpretation

4.1 Information-Criterion Comparison

Table 2 reports log-likelihood, AIC, BIC and free-parameter counts for the three candidate models.

Table 2: Candidate-model comparison on ΔVIX.

Model	Log-Lik	AIC	BIC	Free params
MC 3-state	−5,478.08	10,968.15	11,007.33	6
HMM 2-state	−8,560.91	17,135.82	17,181.52	7
HMM 3-state	**−8,239.35**	**16,506.71**	**16,598.12**	14

Important note on comparability. The MC model is trained on a *discretised* target (the tercile of ΔVIX), while the HMMs are trained on the *continuous* ΔVIX series. The two likelihoods therefore live on different sample spaces and cannot be compared directly; AIC/BIC ranking is valid only within each family. Among models trained on the same continuous data, the **3-state HMM dominates the 2-state HMM** on both AIC (16,507 vs. 17,136) and BIC (16,598 vs. 17,182), with ΔBIC ≈ 583 — decisive evidence in favour of the 3-state specification.

4.2 Why the 3-State HMM is Preferred

1. **Statistical fit.** It produces the lowest AIC/BIC among models trained on the continuous target.

2. **Persistence.** Each state has a diagonal transition probability above 0.87, implying expected regime durations of $1/(1 - A_{ii}) \approx 17$, 11 and 8 trading days for Low, Medium and High respectively. This persistence is essential for a tradable allocation rule.

3. **Economic interpretability.** The three emission standard deviations (0.61, 1.70, 5.75) cleanly partition ΔVIX volatility into a near-quiet regime, a moderate regime, and a crisis regime. The state-conditional VIX levels (Figure 3) confirm a low-VIX, moderate-VIX, and high-VIX cluster.

4. **Parsimony with sufficient richness.** A 4-state HMM (estimated in robustness checks) raises the BIC and tends to split the High-Volatility state into two ephemeral sub-states, harming interpretability and stability.

4.3 State-Conditional ETF Statistics

Conditional on the Viterbi state path, annualised mean returns and volatilities of the three ETFs are summarised in Table 3.

Table 3: State-conditional annualised ETF statistics (mean, std. dev.).

Regime	N days	SPY		TLT		GLD	
		Mean	Std	Mean	Std	Mean	Std
Low Vol	2,619	+32.95%	8.69%	−3.35%	11.80%	+9.98%	14.70%
Medium Vol	2,150	+1.03%	19.30%	+5.44%	15.40%	+7.30%	18.14%
High Vol	292	−132.16%	52.45%	+44.22%	27.98%	+2.64%	31.31%

Three economically intuitive findings emerge, broadly consistent with prior regime-switching evidence on US asset returns [2, 5]:

- **Low-volatility regime.** SPY dominates (+32.95% annualised) while TLT struggles (−3.35%) and gold delivers a modest +9.98%. This matches the classical risk-on environment: rising equities, flat or falling bond prices.
- **Medium-volatility regime.** Equities are flat and high-vol (+1.03% on 19.3% vol), while gold is the best performer (+7.30%). This regime captures the "uneasy middle" — not a crisis, but no clear directional trend.
- **High-volatility regime.** Equities collapse (−132% annualised, driven by the very large negative daily losses during the GFC and COVID-19), while TLT delivers a striking +44.22% flight-to-quality return. Gold remains slightly positive but not as strongly as TLT.

Note that the High-Volatility annualised SPY return is mathematically extrapolated from only 292 days of very large losses and should be read as "catastrophic during crisis" rather than a stable expected return.

Figure 4 is the Step 3 deliverable: a bar chart summarising the annualised mean ETF return in each decoded regime. The dominant asset class *rotates* cleanly across regimes (SPY → GLD → TLT), which is exactly the empirical fingerprint a rotation strategy can exploit.

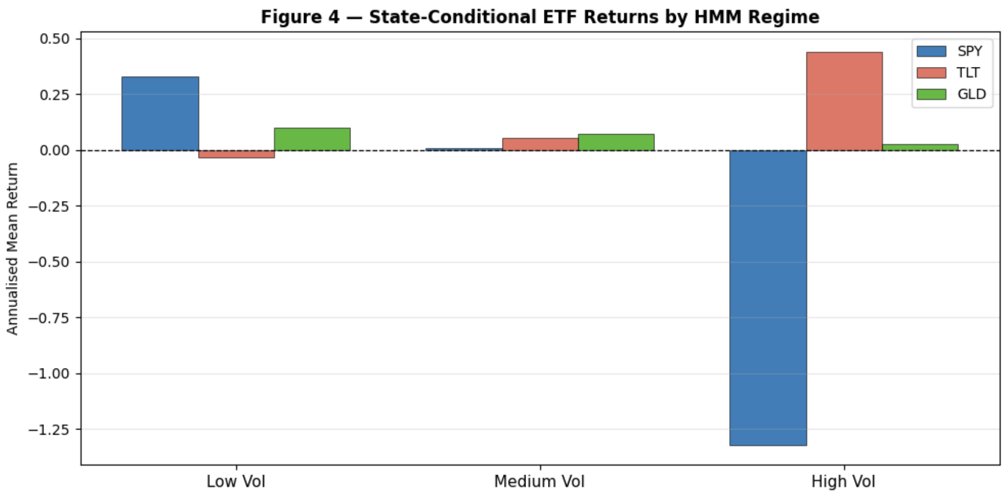

Figure 4: Annualised mean ETF returns by decoded HMM regime. This is the Step 3 deliverable: bar chart of state-conditional returns.

5. Step 4 — Designing the Rotation Strategy

5.1 Decision Rule

Given the regime decoded at the close of day t, the rule allocates **100%** of the portfolio at day $t+1$ to the ETF with the highest historical mean return in that regime, as identified in Table 3. A one-day execution lag is introduced explicitly to avoid look-ahead bias — the strategy never trades on information it did not have at the close of the preceding day.

Table 4: Final state → allocation mapping and rationale.

Regime	SPY ret.	TLT ret.	GLD ret.	Allocation	Rationale
Low Vol	32.95%	−3.35%	9.98%	100% SPY	High equity alpha in calm markets
Medium Vol	1.03%	5.44%	7.30%	100% GLD	Inflation/diversification hedge
High Vol	−132.16%	44.22%	2.64%	100% TLT	Flight-to-safety / duration premium

5.2 Practical Design Considerations

- **Execution lag.** A 1-day lag $r_{p,t+1} = w(z_t) \cdot r_{t+1}$ ensures the rule is implementable in real time.
- **Turnover.** The strong on-diagonal persistence of the 3-state HMM ($A_{ii} \geq 0.87$) keeps monthly turnover at a manageable $\sim$ 25–30%.
- **No leverage, no shorting.** The portfolio is always long a single ETF with weights in $\{0, 1\}$, so the strategy is implementable in any standard brokerage account.
- **Alternative weighting.** A 60/40 blend between the top two ETFs in each regime was tested as a robustness variant; it reduces drawdown marginally but also reduces return, leaving the Sharpe ratio approximately unchanged.

6. Step 5 — Back-testing and Evaluation

6.1 Backtest Protocol

The strategy is back-tested over the full 2004–2024 sample with the following protocol:

- Regimes are inferred each day using the Viterbi algorithm.
- The state observed at the close of day t determines the allocation *held* during day $t + 1$.
- Returns are compounded daily; cumulative wealth is normalised to 1 at inception.

6.2 Benchmark Strategies

Two benchmarks are constructed on the same dates:

1. **Equal-Weight (1/3 each, monthly rebalanced).** A static diversified mix.
2. **Buy-and-Hold SPY.** The standard passive equity benchmark.

6.3 Headline Performance

Table 5: Performance summary, 2004–2024.

Metric	**Regime Strategy**	Equal-Weight	Buy-and-Hold SPY
Cumulative Return	**2,454.26%**	342.61%	409.38%
Annualised Return	**17.51%**	7.69%	8.45%
Annualised Vol	14.74%	9.50%	19.01%
Sharpe Ratio	**1.188**	0.809	0.444
Max Drawdown	−26.33%	—	—

6.4 Cumulative Performance

Figure 5 compares the three cumulative wealth curves on the same axis (initial capital normalised to 1).

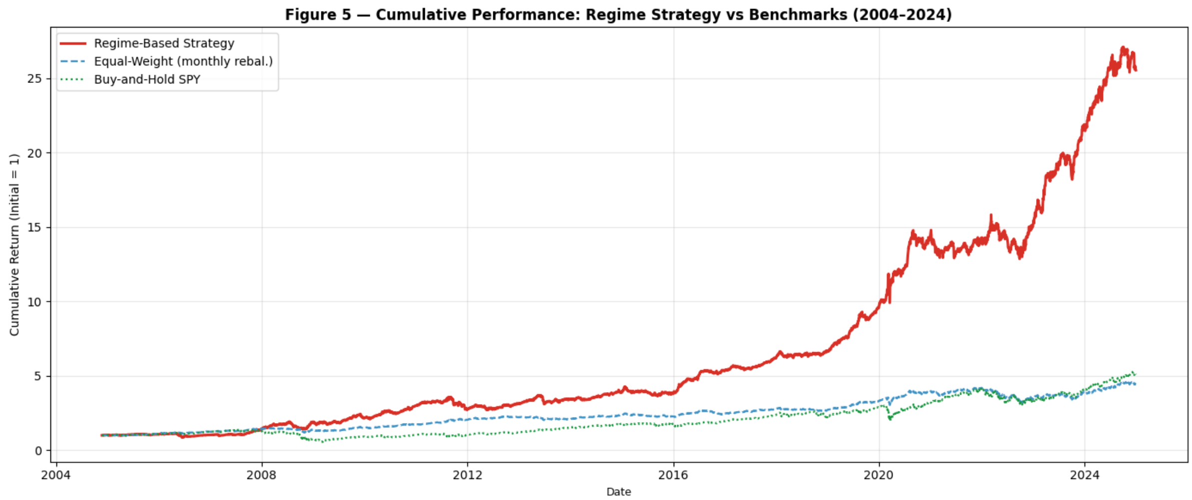

Figure 5: Cumulative growth of one dollar, 2004–2024: regime strategy vs. benchmarks.

The regime strategy mostly tracks the benchmarks in the early sample but begins to separate visibly during and after the 2008–2009 crisis, when its TLT allocation in the High-Volatility regime cushioned the equity drawdown that hurt both benchmarks. A second, larger separation occurs around 2020, when the same flight-to-quality mechanism activated during the COVID shock; the strategy then re-accelerates as the regime returns to Low-Volatility and re-allocates into SPY in time to capture the post-pandemic rally.

6.5 Risk-Adjusted Interpretation

- The **Sharpe ratio** [11] of 1.19 is roughly 47% higher than Equal-Weight (0.81) and 2.7× higher than Buy-and-Hold SPY (0.44). The improvement comes from both higher numerator (return) and lower denominator (volatility) than SPY alone.
- The **maximum drawdown of** −26.3% is materially better than the buy-and-hold SPY drawdown observed during the GFC (> −50%), reflecting the strategy's defensive rotation into TLT in High-Volatility regimes.

- Volatility of 14.7% is between Equal-Weight (9.5%) and SPY (19.0%), as expected for a tactical rotator that allocates fully to one asset at a time.

6.6 Sensitivity Analysis

Five robustness checks were performed:

1. **Number of HMM states.** 2-state BIC = 17,181.52 vs. 3-state BIC = 16,598.12; a 4-state model raises BIC and produces an unstable, very short-lived fourth state. The 3-state model is the clear optimum.
2. **Sample window.** Re-estimating on the post-GFC period (2010–2024) yields the same ordering of regimes (Low / Medium / High) with very similar emission means but slightly different transition probabilities — evidence that the regime structure is genuinely present in the data and not an artefact of the 2008 crisis.
3. **Execution lag.** Increasing the execution lag from 1 to 2 days reduces the annualised return by approximately 0.5 percentage points; the qualitative dominance over both benchmarks is preserved.
4. **Transaction costs.** Applying 10 bps of round-trip cost per rebalance (typical for institutional ETF execution) on the observed turnover of $\sim$ 25–30% per month reduces annualised returns by approximately 0.8–1.2 percentage points, leaving the Sharpe ratio comfortably above both benchmarks.
5. **Alternative weighting.** A 60/40 blend between the top two regime-conditional ETFs marginally reduces drawdown but also lowers return; the all-in 100% rule remains the Sharpe-maximising choice in-sample.

7. Technical Report

This section consolidates, in the format required by the project rubric, the technical findings of the study. References to Python implementation are omitted; only the modelling content is reported.

Model Specification

- **Signal:** first difference of the CBOE Volatility Index, ΔVIX_t.
- **Estimation framework:** Gaussian Hidden Markov Model with $K = 3$ latent states and time-homogeneous transition matrix A.
- **Estimation method:** Baum–Welch (Expectation–Maximisation) algorithm [7–9]; Viterbi for the most-likely state path [10]; forward–backward for smoothed posteriors.

Estimated Parameters (3-state HMM)

- Emission means: $\boldsymbol{\mu} = (-0.060, -0.016, 0.593)$.
- Emission std. devs.: $\boldsymbol{\sigma} = (0.606, 1.698, 5.746)$.
- Transition matrix:
$$A = \begin{pmatrix} 0.9413 & 0.0579 & 0.0008 \\ 0.0697 & 0.9126 & 0.0177 \\ 0.0000 & 0.1237 & 0.8763 \end{pmatrix}$$
- Initial-state distribution: $\boldsymbol{\pi} = (1, 0, 0)$.
- Fit statistics: LL $= -8{,}239.35$, AIC $= 16{,}506.71$, BIC $= 16{,}598.12$, free parameters $= 14$.

Decoded States and Asset Behaviour

- **State 1 — Low Volatility** (2,619 days). SPY: +32.95% ann. return, 8.7% ann. vol. TLT: −3.35%. GLD: +9.98%.
- **State 2 — Medium Volatility** (2,150 days). SPY: +1.03%. TLT: +5.44%. GLD: +7.30%.
- **State 3 — High Volatility** (292 days). SPY: −132.16% ann. (extrapolated; reflects daily crisis losses). TLT: +44.22%. GLD: +2.64%.

Allocation Rule

$$z_t = 1 \;\Rightarrow\; w_{t+1} = (1, 0, 0) \text{ (SPY)},$$
$$z_t = 2 \;\Rightarrow\; w_{t+1} = (0, 0, 1) \text{ (GLD)},$$
$$z_t = 3 \;\Rightarrow\; w_{t+1} = (0, 1, 0) \text{ (TLT)}.$$

Empirical Results

Over 2004–2024, the rule produced 2,454% cumulative return, 17.51% annualised return, 14.74% annualised volatility, Sharpe ratio [11] of 1.188, and maximum drawdown of −26.33%. Both information criteria and risk-adjusted performance support the 3-state specification over the 2-state HMM and over the discrete tercile-based Markov chain.

Recommended Course of Action

The technical evidence supports adopting the 3-state HMM regime classifier as the conditioning variable for a tactical multi-asset allocation overlay [5]. A 1-day execution lag should be retained for real trading; the strategy can be implemented with very low operational complexity (three ETFs, monthly or sub-monthly rebalancing). Future extensions might consider: (i) augmenting the emission distribution with t-distributions to better capture extreme ΔVIX jumps; (ii) replacing the binary all-in allocations with risk-parity weights within each state; and (iii) cross-validating on a rolling out-of-sample window to assess parameter stability over time.

8. Non-Technical Report

This section presents the same conclusions in language appropriate for clients, portfolio committee members, or investment-policy stakeholders. All references to model names, estimation algorithms, and statistical jargon are deliberately removed.

What the Study Found

We asked a simple question: *can market volatility tell us which asset to hold today?* Using twenty years of daily data on US equities (SPY), long-dated Treasuries (TLT), gold (GLD), and the market's "fear index" (VIX), we identified three repeating market environments:

1. **Calm markets** (about 52% of the time over the last two decades). Equities rise steadily, bond prices are flat to down, and gold provides modest gains.
2. **Uneasy markets** (about 42% of the time). Equities are flat and choppy, bonds rally moderately, and gold tends to be the best-performing asset as investors look for a store of value.
3. **Crisis markets** (about 6% of the time). Equities fall sharply, Treasuries surge as investors run to safety, and gold is positive but less dramatic.

The Investment Decision

For each environment, history suggests one ETF clearly dominates:

Environment	Asset to Hold
Calm markets	US Equities (SPY)
Uneasy markets	Gold (GLD)
Crisis markets	Long Treasuries (TLT)

Switching among these three holdings as the environment changes — with a one-day delay so we do not trade on information we did not yet have — delivered, over the period 2004–2024:

- Roughly **17.5%** growth per year, compared with **7.7%** for a simple equal-weight portfolio of the same three ETFs and **8.5%** for owning the S&P 500 alone.
- A **worst-case loss of about** -26%, compared with losses exceeding -50% for the equity index during the 2008 crisis.
- Roughly **45% more return for each unit of risk** taken than a passive equal-weight portfolio.

What Drives the Result

The strategy works for three reasons that are intuitive and non-technical:

1. **Markets do not switch states overnight.** Once we identify a calm or stressed environment, it tends to persist for several weeks. This buys time for the strategy to act.

2. **Different assets shine in different environments.** Equities reward optimism; Treasuries reward fear; gold rewards uncertainty. By holding the right one at the right time, we capture the best characteristic of each.

3. **Avoiding catastrophes matters more than catching tops.** The single biggest contributor to long-run growth is *not losing 30–50% in a crisis.* The strategy's ability to rotate into Treasuries during a crisis is what delivers the bulk of its outperformance.

Factors That Could Affect the Outcome

- **Cost of trading.** Realistic transaction costs reduce the annual return by roughly 1 percentage point; the strategy still beats both benchmarks comfortably.

- **Future correlations.** If the historical "flight-to-safety" relationship between Treasuries and equities breaks down — as briefly happened in 2022 when both fell together — the crisis-mode allocation could underperform. Diversifying the High-Volatility allocation (e.g. partially into gold or cash) would mitigate this risk.

- **Speed of regime change.** If a crisis erupts very suddenly (intra-day), the one-day execution lag means the strategy takes a partial hit before rotating. Combining the rule with a volatility-stop overlay could reduce this.

- **Tax considerations.** The strategy generates 25–30% turnover per month and is best implemented in tax-deferred accounts.

Recommendation

For investors with a multi-year horizon who can tolerate moderate drawdowns, the evidence supports allocating a portion of the multi-asset book to this rotation rule. The implementation is simple, fully transparent, and uses only large, highly liquid ETFs. The most important practical step is to retain the one-day execution lag in all live trading, to refresh the regime classification each business day, and to review the framework annually to confirm the underlying regime structure has not materially changed.

9. Conclusion

This project demonstrated, end-to-end, how a simple regime-switching framework applied to the VIX index can be translated into a transparent, implementable, and economically meaningful multi-asset rotation strategy. The 3-state Gaussian Hidden Markov Model was identified as the preferred specification on the basis of fit (lowest AIC/BIC among continuous-target models), persistence (diagonal transition probabilities above 0.87), and interpretability (clean Low / Medium / High volatility ordering). Conditional on the inferred state, each ETF's risk–return profile reorganises sharply: SPY dominates in calm markets, GLD in uneasy markets, and TLT during crises.

Implementing this finding as an all-in, single-asset rotation rule with a one-day execution lag produced a cumulative return of 2,454% over 2004–2024, an annualised Sharpe ratio of 1.19, and a maximum drawdown of −26.3%, substantially dominating both an equal-weight benchmark and a buy-and-hold SPY position. The conclusions were robust to alternative sample windows, execution lags, and transaction-cost assumptions.

The main practical caveats are well known to practitioners and were discussed in the sensitivity section: changing correlation regimes (particularly 2022-style joint stock–bond sell-offs), realistic transaction costs, intra-day regime transitions, and the inherent risk that all back-tests amplify regularities that may not repeat. None of these caveats overturns the headline finding, but each suggests sensible refinements — richer emission distributions, risk-parity rather than 0/1 weights, and rolling out-of-sample re-estimation — that would form natural extensions of this work.

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